

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12418222
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Kim; Min Gu

Electronic control unit for redundancy of motor driven power steering for vehicle

Abstract

An electronic control unit for redundancy of a motor driven power steering for a vehicle includes: a first motor control circuit unit having a circuit structure to control an operation of a motor; a second motor control circuit unit having a circuit structure to control the operation of the motor, and disposed in parallel to the first motor control circuit unit along an x-direction; and a redundancy control unit to control the second motor control circuit unit to perform a motor control, when an abnormality occurs in the first motor control circuit unit.

Inventors:	Kim; Min Gu (Yongin-si, KR)
Applicant:	HYUNDAI MOBIS CO., LTD. (Seoul, KR)
Family ID:	1000008822286
Assignee:	Hyundai Mobis Co., Ltd. (Seoul, KR)
Appl. No.:	17/724705
Filed:	April 20, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230198348 A1	Jun. 22, 2023

Foreign Application Priority Data

KR	10-2021-0183612	Dec. 21, 2021
----	-----------------	---------------

Publication Classification

Int. Cl.: H02K11/33 (20160101); B60R16/023 (20060101); B62D5/04 (20060101); H02K3/50 (20060101); H02K5/22 (20060101); H02K9/22 (20060101)

U.S. Cl.:

CPC H02K11/33 (20160101); B60R16/0231 (20130101); B62D5/046 (20130101); H02K3/50 (20130101); H02K5/225 (20130101); H02K9/227 (20210101); H02K2203/09 (20130101); H02K2211/03 (20130101); H02K2213/06 (20130101)

Field of Classification Search

CPC: H02K (11/33); H02K (11/30); H02K (11/38); H02K (3/50); H02K (5/225); H02K (9/227); H02K (2203/09); H02K (2213/06); H02K (2211/03); B60R (16/0231); B60R (16/02); B62D (5/046)

USPC: 310/68R; 310/64; 310/71

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2016/0355210	12/2015	Inada	N/A	H05K 7/20854
2018/0178739	12/2017	Fujimoto	N/A	B60R 16/027
2018/0178829	12/2017	Fujimoto	N/A	N/A
2020/0391785	12/2019	Kim	N/A	H02K 11/33

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10-1072749	12/2010	KR	N/A
10-2018-0023997	12/2017	KR	N/A

OTHER PUBLICATIONS

Korean Office Action issued on Oct. 31, 2023, in counterpart Korean Patent Application No. 10-2021-0183612 (4 pages in Korean). cited by applicant

Primary Examiner: Singh; Alexander A

Attorney, Agent or Firm: NSIP Law

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority from and the benefit of Korean Patent Application No. 10-2021-0183612, filed on Dec. 21, 2021, which is hereby incorporated by reference for all purposes as if set forth herein.

BACKGROUND

Field

(2) Exemplary embodiments of the present disclosure relate to an ECU (Electronic Control Unit) for redundancy of an MDPS (Motor Driven Power Steering) for a vehicle, and more particularly, to an ECU for redundancy of an MDPS for a vehicle, which controls the operation of a motor to generate a driving force of an MDPS for a vehicle.

Discussion of the Background

(3) In general, an ECU of an MDPS for a vehicle has a structure in which one motor control circuit unit for controlling the operation of a motor is connected to a vehicle-side power supply, a CAN (Controller Area Network) line, and a torque sensor.

(4) The motor control circuit unit is a main circuit unit of the ECU for controlling the operation of the motor which generates a driving force of the MDPS. When the motor control circuit unit fails, the motor may lose the driving force, and the MDPS may not be normally operated, and may put a driver in danger.

(5) Recently, the development of autonomous vehicles has been conducted with the aim of fully autonomous driving. Thus, various attempts are made to further reinforce the operation stability of the MDPS. However, when the motor control circuit unit fails, the conventional MDPS has no device capable of taking place of the motor control circuit unit. Thus, the conventional MDPS has limitations in securing the stability of autonomous driving.

(6) Therefore, there is a need for a device capable of solving the problem.

(7) The related art of the present disclosure is disclosed in Korean Patent No. 10-1072749 registered on Oct. 5, 2011 and entitled “Electronic Control Unit of Electric Power Steering Apparatus.”

SUMMARY

(8) This Summary is provided to introduce a selection of concepts in simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(9) In one general aspect, an ECU (Electronic Control Unit) for redundancy of an MDPS (Motor Driven Power Steering) for a vehicle includes: a first motor control circuit unit having a circuit structure configured to control an operation of a motor; a second motor control circuit unit having a circuit structure configured to control the operation of the motor, and disposed in parallel to the first motor control circuit unit in an x-direction; and a redundancy control unit configured to control the second motor control circuit unit to perform a motor control, when an abnormality occurs in the first motor control circuit unit.

(10) The first motor control circuit unit may include: a first circuit board having a circuit structure configured to independently control the operation of the motor; a first power terminal connection part disposed in a first area of the first circuit board, and connected to a power terminal part connected to a vehicle-side power connector to receive power; a first communication terminal connection part disposed in a second area of the first circuit board, and connected to a CAN terminal part connected to a vehicle-side CAN connector to receive a signal; and a first sensor terminal connection part disposed in a third area of the first circuit board, and connected to a torque sensor terminal part connected to a vehicle-side torque sensor connector to receive a signal.

(11) The second motor control circuit unit may include: a second circuit board having a circuit structure configured to independently control the operation of the motor; a second power terminal connection part disposed on an x-direction extension line of a first area of the second circuit board, and connected to the power terminal part to receive power; a second communication terminal connection part disposed on an x-direction extension line of a second area of the second circuit board, and connected to the CAN terminal part to receive a signal; and a second sensor terminal connection part disposed on an x-direction extension line of a third area of the second circuit board, and connected to the torque sensor terminal part to receive a signal.

(12) The redundancy control unit may include: a circuit detection part configured to detect an operation state of the first motor control circuit unit; an abnormality determination part configured to determine whether the first motor control circuit unit is abnormal, according to a signal detected by the circuit detection part; and a driving circuit switching part configured to stop the operation of the motor by the first motor control circuit unit, and perform circuit switching such that the second motor control circuit unit controls the operation of the motor, when the abnormality determination part determines that the first motor control circuit unit is abnormal, while the first motor control circuit unit controls the motor.

(13) The circuit detection part may include: a first circuit detector configured to detect the operation state of the first motor control circuit unit; and a second circuit detector configured to detect an operation state of the second motor control circuit unit, wherein the abnormality determination part is configured to determine whether the first circuit detector is abnormal, by comparing detection signals of the first and second circuit detectors or by comparing the detection signal of the first circuit detector to a preset value.

(14) The ECU may further include a motor coupling unit coupled to a motor housing, and disposed between the first and second motor control circuit units.

(15) The motor coupling unit may include: a board support part fixed to the first and second motor control circuit units, and configured to receive heat radiated from the first and second motor control circuit units; and a heat radiation support part connected to the board support part, and configured to radiate heat received from the board support part.

(16) The board support part may include: a support body part; a first board support part disposed on a first surface of the support body part, such that the first motor control circuit unit abuts the first board support part; a second board support part disposed on a second surface of the support body part, such that the second motor control circuit unit abuts the second board support part; and a terminal through-hole disposed through the support body part, such that a power terminal part connected to a vehicle-side power connector, a CAN terminal part connected to a vehicle-side CAN connector, and a torque sensor terminal part connected to a vehicle-side torque sensor connector are extended through the terminal through-hole in the x-direction.

(17) The board support part may include: a support body part; a contact support part disposed on a first surface of the support body part, such that one of the first and second motor control circuit units abuts the contact support part; and a spacing support part protruding from the contact support part in the x-direction, and configured to support the other of the first and second motor control circuit units.

(18) The ECU may further include a double-layer power terminal part having a first side connected to a vehicle-side power connector and a second side diverging into double layers, extended in the x-direction, and connected to the first motor control circuit unit and the second motor control circuit unit.

(19) The double-layer power terminal part may include: an external power connection part connected to the vehicle-side power connector; a first-layer power connection part extended from the external power connection part in the x-direction, and connected to one of the first and second motor control circuit units; and a second-layer power connection part extended in the x-direction from a position spaced apart from the first-layer power connection part, and connected to the other of the first and second motor control circuit units.

(20) The first-layer power connection part may protrude in the x-direction from a first side of the external power connection part, and have a first end connected to the first motor control circuit unit. The second-layer power connection part may include: a power diverging part formed on a second side of the external power connection part to protrude in one of a y-direction and a z-direction; a power middle extension part extended in the other of the y-direction and the z-direction from the power diverging part; and a power extension end part extended in the x-direction from the power middle extension part, and having an end spaced apart from the first end of the first-layer

power connection part in the x-direction, the y-direction and the z-direction.

(21) The double-layer power terminal part may include: a first bus bar having the external power connection part, the first-layer power connection part and the second-layer power connection part; a second bus bar disposed symmetrically with the first bus bar in a y-direction; and an intersection space defined between the first and second bus bars, such that at least a part of a CAN terminal part connected to a vehicle-side CAN connector is disposed in the intersection space.

(22) The ECU may further include a double-layer CAN terminal part having a first side connected to a vehicle-side CAN connector and a second side diverging into double layers, extended in the x-direction, and connected to the first and second motor control circuit units. The double-layer CAN terminal part may include: an external CAN connection part connected to the vehicle-side CAN connector; a second-layer CAN connection part extended in the x-direction from the external CAN connection part, and connected to one of the first and second motor control circuit units; and a first-layer CAN connection part diverging in a y-direction or a z-direction from the external CAN connection part and extended in the x-direction, and connected to the other of the first and second motor control circuit units.

(23) The second-layer CAN connection part may be formed on a first side of the external CAN connection part to extend in the x-direction, and have a first end connected to the second motor control circuit unit. The first-layer CAN connection part may include: a CAN diverging part disposed on a second side of the external CAN connection part to protrude in one of the y-direction and the z-direction; a CAN middle extension part extended in the other of the y-direction and the z-direction from the CAN diverging part; and a CAN extension end part extended in the x-direction from the CAN middle extension part, and having a first end spaced apart from the first end of the second-layer CAN connection part in the x-direction, the y-direction and the z-direction.

(24) The ECU may further include a double-layer sensor terminal part having a first side connected to a vehicle-side torque sensor connector and a second side diverging into double layers, extended in the x-direction, and connected to the first and second motor control circuit units. The double-layer sensor terminal part may include: an external sensor connection part connected to the vehicle-side torque sensor connector; a first-layer sensor connection part diverging in the x-direction from the external sensor connection part, and connected to one of the first and second motor control circuit units; and a second-layer sensor connection part diverging in a y-direction or z-direction from the external sensor connection part and extended in the x-direction, and connected to the other of the first and second motor control circuit units.

(25) The first-layer sensor connection part may be formed on a first side of the external sensor connection part to extend in the x-direction, and have a first end connected to the first motor control circuit unit. The second-layer sensor connection part may include: a sensor diverging part disposed on a second side of the external sensor connection part to protrude in one of the y-direction and the z-direction; a sensor middle extension part extended in the other of the y-direction and the z-direction from the sensor diverging part; and a sensor extension end part extended in the x-direction from the sensor middle extension part, and having a first end spaced apart from the first end of the first-layer sensor connection part in the x-direction, the y-direction and the z-direction.

(26) In another general aspect, an ECU for redundancy of an MDPS for a vehicle may include: a motor coupling unit coupled to a motor; a first motor control circuit unit installed on the motor coupling unit; a second motor control circuit unit installed on the motor coupling unit, and disposed in parallel to the first motor control circuit unit along an x-direction; a power connector having a first side connected to a vehicle-side power connector and a second side connected to the first and second motor control circuit units; and a redundancy control unit configured to control the second motor control circuit unit to perform a motor control, when an abnormality occurs in the first motor control circuit unit.

(27) The power connector may include: a double-layer power terminal part having a bus bar structure in which a first side thereof is connected to the vehicle-side power connector and a second

side thereof diverges into double layers, extended in the x-direction, and connected to the first and second motor control circuit units; a CAN terminal part connected to a vehicle-side CAN connector; and a first insert molding part configured to connect the double-layer power terminal part to the CAN terminal part through insert molding, and installed on the motor coupling part.

(28) The CAN terminal part may include a double-layer CAN terminal part having a bus bar structure in which a first side thereof is connected to the vehicle-side CAN connector and a second side thereof diverges into double layers, extended in the x-direction, and connected to the first and second motor control circuit units. At least a part of a first area in which the double layer power terminal part is disposed and at least a part of a second area in which the double-layer CAN terminal part is disposed may cross or overlap each other.

(29) The ECU may further include a torque sensor connector connected to a vehicle-side torque sensor connector. The torque sensor connector may include: a double-layer sensor terminal part having a bus bar structure in which a first side thereof is connected to the vehicle-side torque sensor connector and a second side thereof diverges into double layers, extended in the x-direction, and connected to the first and second motor control circuit units; and a second insert molding part formed by insert molding the double-layer sensor terminal part, and fitted and installed into the motor coupling unit in the x-direction.

(30) Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an assembled view illustrating main parts of an ECU (Electronic Control Unit) for redundancy of an MDPS (Motor Driven Power Steering) for a vehicle in accordance with an embodiment of the present disclosure.
- (2) FIG. 2 is a cross-sectional view illustrating the main parts of the ECU for redundancy of an MDPS for a vehicle in accordance with the embodiment of the present disclosure.
- (3) FIG. 3 is a cross-sectional view illustrating the main parts of the ECU for redundancy of the MDPS for a vehicle in accordance with an embodiment of the present disclosure, when seen in a different direction from FIG. 2.
- (4) FIG. 4 is an exploded perspective view illustrating the main parts of the ECU for redundancy of an MDPS for a vehicle in accordance with the embodiment of the present disclosure.
- (5) FIG. 5 is a perspective view illustrating the main parts of the ECU for redundancy of an MDPS for a vehicle in accordance with the embodiment of the present disclosure.
- (6) FIG. 6 is a plan view of FIG. 5.
- (7) FIG. 7 is a perspective view schematically illustrating a double-layer power terminal part in accordance with the embodiment of the present disclosure.
- (8) FIG. 8 is a perspective view schematically illustrating a double-layer CAN (Controller Area Network) terminal part in accordance with the embodiment of the present disclosure.
- (9) FIG. 9 is a perspective view schematically illustrating a double-layer sensor terminal part in accordance with the embodiment of the present disclosure.
- (10) FIG. 10 is a cross-sectional view illustrating main parts of an ECU for redundancy of an MDPS for a vehicle in accordance with another embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

(11) Hereinafter, an ECU (Electronic Control Unit) for redundancy of an MDPS (Motor Driven Power Steering) for a vehicle will be described below with reference to the accompanying drawings through various exemplary embodiments. It should be noted that the drawings are not to precise scale and may be exaggerated in thickness of lines or sizes of components for descriptive

convenience and clarity only. Furthermore, the terms as used herein are defined by taking functions of the invention into account and can be changed according to the custom or intention of users or operators. Therefore, definition of the terms should be made according to the overall disclosures set forth herein.

(12) FIG. 1 is an assembled view illustrating main parts of an ECU for redundancy of an MDPS for a vehicle in accordance with an embodiment of the present disclosure, FIG. 2 is a cross-sectional view illustrating the main parts of the ECU for redundancy of an MDPS for a vehicle in accordance with the embodiment of the present disclosure, and FIG. 3 is a cross-sectional view illustrating the main parts of the ECU for redundancy of the MDPS for a vehicle in accordance with an embodiment of the present disclosure, when seen in a different direction from FIG. 2.

(13) Referring to FIGS. 1 to 3, an ECU 1 for redundancy of an MDPS for a vehicle in accordance with an embodiment of the present disclosure includes a housing 10, a motor coupling unit 20, a first motor control circuit unit 30, a second motor control circuit unit 40, a power connector 50, a torque sensor connector 60 and a redundancy control unit 70.

(14) The housing 10 is coupled to one side of the motor coupling unit 20, and covers the first motor control circuit unit 30, the second motor control circuit unit 40, the power connector 50, the torque sensor connector 60 and the like, which are mounted on the motor coupling unit 20. In the following descriptions of the present disclosure, 'the inside of the housing 10' indicates a space formed between the housing 10 and the motor coupling unit 20.

(15) Furthermore, the term 'vehicle side' written with a component in the descriptions of the present disclosure indicates that the component is disposed outside the housing 10. In the descriptions of the present disclosure, a component written with the term 'vehicle-side' is distinguished from the components of the ECU 1 for redundancy for an MDPS for a vehicle in accordance with the embodiment of the present disclosure.

(16) The housing 10 has through-holes 11 formed at positions corresponding to the power connector 50 and the torque sensor connector 60, respectively. The power connector 50 and the torque sensor connector 60, located inside the housing 10, may be connected to a vehicle-side power connector 2, a vehicle-side CAN (Controller Area Network) connector 3 and a vehicle-side torque sensor connector 4, which are located outside of the housing 10, through the through-holes 11.

(17) Hereafter, the positions and directions of the components of the ECU 1 for redundancy for an MDPS for a vehicle in accordance with the embodiment of the present disclosure will be described on the basis of an x-direction, a y-direction and a z-direction, which are illustrated in FIGS. 1 to 3, for convenience of description. The x-direction, the y-direction and the z-direction each include a positive (+) direction and a negative (-) direction, and the positive (+) direction and the negative (-) direction are separately marked only in case of necessity.

(18) The motor coupling unit 20 is a device part coupled to a motor 5, has a flat cylindrical shape corresponding to an opening (with no reference numeral) of a motor housing 6 as a whole, and is coupled to the motor housing 6. The motor coupling unit 20 is made of a metallic material capable of implementing a heat radiation function, and serves to receive heat radiated from the first and second motor control circuit units 30 and 40 and discharge the received heat to the outside of the housing 10.

(19) Each of the first and second motor control circuit units 30 and 40 has a circuit structure capable of independently controlling the operation of the motor 5, and is installed on the motor coupling unit 20. The first and second motor control circuit units 30 and 40 are disposed in parallel to each other with a preset interval provided therebetween in the x-direction.

(20) The first motor control circuit unit 30 has a circuit structure which can reliably control the operation of the motor 5 by itself, without connection to the second motor control circuit unit 40 or even though the second motor control circuit unit 40 completely loses its function. The second motor control circuit unit 40 also has a circuit structure which can reliably control the operation of

the motor **5** by itself, without connection to the first motor control circuit unit **30** or even though the first motor control circuit unit **30** completely loses its function.

(21) That is, the first and second motor control circuit units **30** and **40** each do not indicate parts of a circuit required for controlling the operation of the motor **5**, for example, a power supply unit, a motor control unit and a signal transfer unit, but indicate a circuit capable of fully controlling the operation of the motor **5** by itself, without connection to another motor control circuit unit. The first motor control circuit unit **30** in accordance with the embodiment of the present disclosure is configured as one circuit board. If necessary, however, the first motor control circuit unit **30** may have a structure divided into a plurality of circuit boards. The second motor control circuit unit **40** is also configured as one circuit board, but may have a structure divided into a plurality of circuit boards.

(22) The power connector **50** has one side connected to the vehicle-side power connector **2** and the other side connected to the first and second motor control circuit units **30** and **40** disposed in parallel to each other in the x-direction. When the vehicle-side power connector **2** and the vehicle-side CAN connector **3** are configured as one part, the power connector **50** has a structure which can be connected to each of the vehicle-side power connector **2** and the vehicle-side CAN connector **3**.

(23) The torque sensor connector **60** has one side connected to the vehicle-side torque sensor connector **4** and the other side connected to the first and second motor control circuit units **30** and **40** disposed in parallel to each other in the x-direction. The vehicle-side torque sensor connector **4** is connected to a torque sensor (not illustrated) which is installed to sense a steering torque or steering angle of a steering wheel. The torque sensor connector **60** may be provided to control the operation of the motor **5** in connection with the torque sensor.

(24) The redundancy control unit **70** is a device part for implementing redundancy for controlling the operation of the motor **5**, and allows the second motor control circuit unit **40** to control the operation of the motor **5**, when an abnormality occurs in the first motor control circuit unit **30**. Hereafter, between the first and second motor control circuit units **30** and **40**, a motor control circuit unit which is substantially controlling the operation of the motor **5** will be referred to as 'an enabled circuit unit', for convenience of description.

(25) When an abnormality of the first motor control circuit unit **30** is sensed while the first motor control circuit unit **30** is set and operated as the enabled circuit unit, the redundancy control unit **70** stops the operation of the motor **5**, which has been controlled by the first motor control circuit unit **30**, and simultaneously allows the second motor control circuit unit **40** to control the operation of the motor **5** in place of the first motor control circuit unit **30**. In other words, the redundancy control unit **70** switches the enabled circuit unit to the second motor control circuit unit **40**.

(26) Referring to FIGS. **2** to **3**, the motor coupling unit **20** in accordance with the embodiment of the present disclosure includes a board support part **21** and a heat radiation support part **28**.

(27) The board support part **21** covers the opening of the motor housing **6**, and simultaneously supports the first and second motor control circuit units **30** and **40** at preset positions. The first and second motor control circuit units **30** and **40** are installed on the board support part **21** with a preset interval provided therebetween in the x-direction. The board support part **21** in accordance with the embodiment of the present disclosure includes a support body part **22**, a first board support part **23**, a second board support part **24** and a terminal through-hole **25**.

(28) The support body part **22** is a part forming the basic frame of the board support part **21**, and has a disk shape with widths in the y- and z-directions as a whole. This structure considers that the motor housing **6** has a circular opening, but the shape of the support body part **22** is not specifically limited. The shape of the support body part **22** may be changed in various manners depending on the shapes and structures of the motor housing **6** and the first and second motor control circuit units **30** and **40**.

(29) The first board support part **23** is formed on one side of the support body part **22** in the x-direction, which faces the housing **10**, such that the first motor control circuit unit **30** abuts on the

first board support part **23**. The second board support part **24** is formed on the other surface of the support body part **22** in the x-direction, which faces the motor **5**, such that the second motor control circuit unit **40** abuts on the second board support part **24**.

(30) With the first and second motor control circuit units **30** and **40** installed on the first and second board support parts **23** and **24**, respectively, the first and second motor control circuit units **30** and **40** are disposed in parallel to each other in the x-direction with an interval provided therebetween, the interval corresponding to the x-direction thickness of the support body part **22**. The edge of the one surface of the support body part **22** in the x-direction, which faces the housing **10**, is coupled to the housing **10**.

(31) As the first and second motor control circuit units **30** and **40** abut on both surfaces of the support body part **22** in the x-direction, heats generated from the first and second motor control circuit units **30** and **40** are directly transferred to the support body part **22**. Thus, an excellent heat radiation effect may be stably implemented for both of the first and second motor control circuit units **30** and **40**.

(32) The terminal through-hole **25** serves to provide a passage where the power connector **50** and the torque sensor connector **60** can be extended through the support body part **22** in the x-direction. The terminal through-hole **25** is formed at a position of the support body part **22**, which corresponds to the power connector **50** and the torque sensor connector **60**, so as to pass through the support body part **22** in the x-direction.

(33) A power terminal part **51** and a CAN terminal part **54**, which are included in the power connector **50**, and a torque sensor terminal part **61** included in the torque sensor connector **60** may be connected to both of the first and second motor control circuit units **30** and **40** disposed on both surfaces of the support body part **22** in the x-direction, while extended through the support body part **22** in the x-direction via the terminal through-hole **25**. The power terminal part **51**, the CAN terminal part **54** and the torque sensor terminal part **61** will be described below.

(34) The heat radiation support part **28** protrudes from the edge of the other surface of the board support part **21** in the x-direction, which faces the motor **5**, and is coupled to the motor housing **6**. The motor coupling unit **20** is made of a metallic material capable of implementing a heat transfer function and a heat radiation function, and serves to receive heat radiated from the first and second motor control circuit units **30** and **40** through the board support part **21**, and discharge the received heat to the outside of the housing **10** and the motor housing **6** through the heat radiation support part **28**.

(35) FIG. **4** is an exploded perspective view illustrating the main parts of the ECU for redundancy of an MDPS for a vehicle in accordance with the embodiment of the present disclosure, FIG. **5** is a perspective view illustrating the main parts of the ECU for redundancy of an MDPS for a vehicle in accordance with the embodiment of the present disclosure, and FIG. **6** is a plan view of FIG. **5**.

(36) Referring to FIGS. **4** to **6**, the first motor control circuit unit **30** in accordance with the embodiment of the present disclosure includes a first circuit board **31**, a first power terminal connection part **32**, a first communication terminal connection part **33** and a first sensor terminal connection part **34**.

(37) The first circuit board **31** has a circuit structure which can independently control the operation of the motor **5**. The first circuit board **31** is connected to the power terminal part **51** disposed in a first area **A1** and connected to the vehicle-side power connector **2**, and receives power. The first communication terminal connection part **33** is connected to the CAN terminal part **54** disposed in a second area **A2** of the first circuit board **31** and connected to the vehicle-side CAN connector **3**, and receives a signal.

(38) The first sensor terminal connection part **34** is connected to the torque sensor terminal part **61** disposed in a third area **A3** of the first circuit board **31** and connected to the vehicle-side torque sensor connector **4**, and receives a signal. In the descriptions of the present disclosure, it is assumed that the first area **A1**, the second area **A2** and the third area **A3** each have the cross-sectional shape

of a figure with no concave portion, such as a rectangle, triangle or circle, the figure being completed by extending a line in one direction corresponding to the clockwise direction or the counterclockwise direction.

(39) Referring to FIGS. 4 and 5, the second motor control circuit unit **40** in accordance with the embodiment of the present disclosure includes a second circuit board **41**, a second power terminal connection part **42**, a second communication terminal connection part **43** and a second sensor terminal connection part **44**.

(40) The second circuit board **41** has a circuit structure for independently controlling the operation of the motor **5**. The second power terminal connection part **42** is disposed on an x-direction extension line of the first area **A1** of the second circuit board **41**, and connected to the power terminal part **51** so as to receive power. The second communication terminal connection part **43** is disposed on an x-direction extension line of the second area **A2** of the second circuit board **41**, and connected to the CAN terminal part **54** so as to receive a signal.

(41) The second sensor terminal connection part **44** is disposed on an x-direction extension line of the third area **A3** of the second circuit board **41**, and connected to the torque sensor terminal part **61** so as to receive a signal. Hereafter, for convenience of description, the first area **A1** and an area belonging to the x-direction extension line of the first area **A1** are not separately distinguished from each other, but collectively referred to as the first area **A1**. For example, the first area **A1** on the second circuit board **41** indicates an area belonging to the x-direction extension line of the first area **A1**.

(42) FIG. 7 is a perspective view schematically illustrating a double-layer power terminal part in accordance with the embodiment of the present disclosure.

(43) Referring to FIG. 4, the power connector **50** in accordance with the embodiment of the present disclosure includes the power terminal part **51**, the CAN terminal part **54** and a first insert molding part **56**.

(44) The power terminal part **51** has a bus bar structure extended in the x-direction, and is connected to the vehicle-side power connector **2**, the first motor control circuit unit **30** and the second motor control circuit unit **40**. One side of the power terminal part **51** in the x-direction is connected to the vehicle-side power connector **2**, and the other side of the power terminal part **51** in the x-direction is connected to the first and second motor control circuit units **30** and **40**.

(45) In particular, when the other side of the power terminal part **51** in the x-direction is extended in the x-direction, the other side of the power terminal part **51** diverges into a plurality of portions which are connected to the first power terminal connection part **32** of the first motor control circuit unit **30** at a first layer and the second power terminal connection part **42** of the second motor control circuit unit **40** at a second layer. In the descriptions of the present disclosure, such a part having an end portion diverging into double layers is referred to as a double-layer power terminal part **52**.

(46) That is, the power terminal part **51** in accordance with the present disclosure includes the double-layer power terminal part **52** having a bus bar structure that extends in the x-direction while diverging into a plurality of portions which can be simultaneously connected to the first and second motor control circuit units **30** and **40** disposed in parallel to each other in the x-direction. Referring to FIGS. 5 and 7, the double-layer power terminal part **52** in accordance with the embodiment of the present disclosure includes a first bus bar **52A**, a second bus bar **52B**, and an intersection space **53**.

(47) The first and second bus bars **52A** and **52B** are terminals connected to negative and positive terminals of the vehicle-side power connector **2**, respectively, and are disposed symmetrically with each other in the y-direction. Between the first and second bus bars **52A** and **52B**, the intersection space **53** is formed, in which at least a part of the CAN terminal part **54** can be disposed.

(48) When at least a part of a double-layer CAN terminal part **55** is disposed in the intersection space **53**, the intersection space **53** is shared by the first area **A1** in which the first and second

power terminal connection parts **32** and **42** are disposed and the second area **A2** in which the first and second communication terminal connection parts **33** and **43** are disposed. In other words, at least a part of the first area **A1** in which the double-layer power terminal part **52** is disposed and at least a part of the second area **A2** in which the double-layer CAN terminal part **55** is disposed cross or overlap each other.

(49) In accordance with the embodiment of the present disclosure, the first and second areas **A1** and **A2** are not disposed independently of each other, but share an area corresponding to the intersection space **53**. Thus, the double-layer power terminal part **52** and the double-layer CAN terminal part **55**, each of which has a plurality of diverging portions and is formed in a 3D shape, may be efficiently disposed in a limited space. Referring to FIGS. 5 and 7, the double-layer power terminal part **52** in accordance with the embodiment of the present disclosure includes an external power connection part **521**, a first-layer power connection part **522** and a second-layer power connection part **526**.

(50) The external power connection part **521** is a terminal connected to the vehicle-side power connector **2**, and the first-layer power connection part **522** and the second-layer power connection part **526** are terminals connected to the first power terminal connection part **32** and the second power terminal connection part **42**, respectively. The external power connection part **521** has a first width in the y-direction and extends in the x-direction, and the first and second power terminal connection parts **32** and **42** both have a second width smaller than the first width, and diverge from the external power connection part **521** and extend in the x-direction.

(51) As the external power connection part **521** has the first width, it is possible to clearly and stably secure a contact area with the vehicle-side power connector **2**. As the first and second power terminal connection parts **32** and **42** have the second width smaller than the first width, the first and second power terminal connection parts **32** and **42** may be stably connected to the first-layer power connection part **522** and the second-layer power connection part **526**, respectively, without interference with other circuit parts constituting the first and second motor control circuit units **30** and **40**.

(52) The first-layer power connection part **522** diverges so as to protrude in the +x-direction from a +x-direction end portion of the external power connection part **521**, and connects to the first power terminal connection part **32**. The second-layer power connection part **526** diverges so as to protrude in the +x-direction from the external power connection part **521**, and extends in the +x-direction from a position spaced apart in the x-direction from the first-layer power connection part **522**.

(53) More specifically, the second-layer power connection part **526** diverges so as to protrude in the y-direction from an x-direction middle portion of the external power connection part **521**, passes through the terminal through-hole **25** of the motor coupling unit **20** in the +x-direction, and then connects to the second power terminal connection part **42**. Referring to FIG. 7, the second-layer power connection part **526** of the first bus bar **52A** in accordance with the embodiment of the present disclosure includes a power diverging part **527**, a power middle extension part **528** and a power extension end part **529**.

(54) The power diverging part **527** protrudes in the +y-direction from a position corresponding to the x-direction middle portion and a +y-direction end portion of the external power connection part **521**. The power middle extension part **528** is extended in the +z-direction from a +x-direction end portion of the power diverging part **527**. The power extension end part **529** is extended in the +x-direction from a +z-direction end portion of the power middle extension part **528**.

(55) The +x-direction end portion of the power extension end part **529** is spaced apart in the x-, y- and z-directions from the +x-direction end portion of the first-layer power connection part **522**. That is, the +x-direction end portions of the first-layer power connection part **522** and the power extension end part **529** are clearly spaced apart from each other in all directions corresponding to the x-, y- and z-directions.

(56) Therefore, it is possible to lower the possibility that the first-layer power connection part **522**

and the second-layer power connection part **526** will be misassembled or interfere with each other during an assembling or manufacturing process, due to an error or mistake, thereby further improving the productivity. Furthermore, it is possible to minimize the possibility that the damage to one power connection part will have an influence on the other power connection part, thereby securing structural stability.

(57) The double-layer power terminal part **52** having the above-described structure may be easily manufactured as one body through a simple process of cutting a metallic material into a structure including the external power connection part **521**, the first-layer power connection part **522** and the second-layer power connection part **526**, and then bending portions corresponding to the power middle extension part **528** and the power extension end part **529** in a multi-stage manner in the +z-direction and the +x-direction, respectively.

(58) The second bus bar **52B** is located in the -y-direction of the first bus bar **52A**, and the second-layer power connection part **526** of the second bus bar **52B** has a symmetrical shape with the second-layer power connection part **526** of the first bus bar **52A** in the y-direction. The intersection space **53** in accordance with the embodiment of the present disclosure is formed between the second-layer power connection part **526** of the first bus bar **52A** and the second-layer power connection part **526** of the second bus bar **52B**.

(59) FIG. **8** is a perspective view schematically illustrating a double-layer CAN terminal part in accordance with an embodiment of the present disclosure.

(60) Referring to FIGS. **5** and **8**, the CAN terminal part **54** has a bus bar structure extended in the x-direction, and is connected to the vehicle-side CAN connector **3**, the first motor control circuit unit **30** and the second motor control circuit unit **40**. One side of the CAN terminal part **54** in the x-direction is connected to the vehicle-side power connector **2**, and the other side of the CAN terminal part **54** in the x-direction is connected to the first and second motor control circuit units **30** and **40**.

(61) In particular, when the other side of the CAN terminal part **54** in the x-direction is extended in the x-direction, the other side of the CAN terminal part **54** diverges into a plurality of portions which are connected to the first communication terminal connection part **33** of the first motor control circuit unit **30** at the first layer and the second communication terminal connection part **43** of the second motor control circuit unit **40** at the second layer. In the descriptions of the present disclosure, such a part having an end portion diverging into double layers is referred to as the double-layer CAN terminal part **55**.

(62) That is, the CAN terminal part **54** in accordance with the present disclosure includes the double-layer CAN terminal part **55** having a bus bar structure that is extended in the x-direction while diverging into a plurality of portions which can be simultaneously connected to the first and second motor control circuit units **30** and **40** disposed in parallel to each other in the x-direction. Furthermore, the CAN terminal part **54** has a structure in which a plurality of double-layer CAN terminal parts **55** are aligned in one direction. In the embodiment of the present disclosure, the plurality of double-layer CAN terminal parts **55** are aligned in the y-direction. The double-layer CAN terminal part **55** in accordance with the embodiment of the present disclosure includes an external CAN connection part **551**, a first-layer CAN connection part **552** and a second-layer CAN connection part **556**.

(63) The external CAN connection part **551** is a terminal connected to the vehicle-side CAN connector **3**, and the first-layer CAN connection part **552** and the second-layer CAN connection part **556** are terminals connected to the first communication terminal connection part **33** and the second communication terminal connection part **43**, respectively. The external CAN connection part **551** is disposed in the +z-direction of the double-layer power terminal part **52**, and extended in the x-direction. The first-layer CAN connection part **552** and the second-layer CAN connection part **556** diverge from the external CAN connection part **551**, and extend in the x-direction.

(64) The first-layer CAN connection part **552** diverges so as to protrude in the -y-direction from a

−y-direction end portion of the external CAN connection part **551**, extends in the −z-direction, and then extends in the +x-direction. An end portion of the first-layer CAN connection part **552**, extending in the +x-direction, passes through the terminal through-hole **25** of the motor coupling unit **20** in the +x-direction, and then connects to the first communication terminal connection part **33**.

(65) The second-layer CAN connection part **556** diverges in a shape to further extend in the +x-direction from a +x-direction end portion of the external CAN connection part **551**, and connects to the second communication terminal connection part **43**. Referring to FIG. **8**, the first-layer CAN connection part **552** includes a CAN diverging part **553**, a CAN middle extension part **554**, and a CAN extension end part **555**.

(66) The CAN diverging part **553** protrudes in the −y-direction from a position corresponding to a +x-direction middle portion and the −y-direction end portion of the external CAN connection part **551**. The CAN middle extension part **554** extends in the −z-direction from the CAN diverging part **553**. The CAN extension end part **555** extends in the +x-direction from a −z-direction end portion of the CAN middle extension part **554**.

(67) The +x-direction end portion of the CAN extension end part **555** is spaced apart in the x-, y- and z-directions from the +x-direction end portion of the second-layer CAN connection part **556**. That is, the +x-direction end portions of the first-layer CAN connection part **552** and the second-layer CAN connection part **556** are clearly spaced apart from each other in all directions corresponding to the x-, y- and z-directions.

(68) Therefore, it is possible to lower the possibility that the first-layer CAN connection part **552** and the second-layer CAN connection part **556** will be misassembled or interfere with each other during an assembling or manufacturing process, due to an error or mistake, thereby further improving the productivity. Furthermore, it is possible to minimize the possibility that the damage to one CAN connection part will have an influence on the other CAN connection part, thereby securing structural stability.

(69) The double-layer CAN terminal part **55** having the above-described structure may be easily manufactured as one body through a simple process of cutting a metallic material into a structure including the external CAN connection part **551**, the first-layer CAN connection part **552** and the second-layer CAN connection part **556**, and then bending a portion corresponding to the CAN middle extension part **554** in the −z-direction.

(70) The double-layer CAN terminal part **55** in accordance with the embodiment of the present disclosure has a structure in which a plurality of double-layer CAN terminal parts **55** are arranged in the y-direction. The external CAN connection part **551** and the second-layer CAN connection part **556** of the double-layer CAN terminal part **55** are located outside the first area **A1** in the +z-direction, and the first-layer CAN connection part **552** of the double-layer CAN terminal **55** is located so as to overlap the inside of the first area **A1** or located in the intersection space **53**.

(71) The first insert molding part **56** is made of an insulating material, and the plurality of double-layer power terminal parts **52**, each including the first and second bus bars **52A** and **52B** arranged symmetrically in the y-direction, and the plurality of double-layer CAN terminal parts **55** arranged in the y-direction, are connected as one body by the first insert molding part **56** through insert molding.

(72) The double-layer power terminal part **52** and the double-layer CAN terminal part **55** become one part corresponding to the power connector **50** by the first insert molding part **56**. As the first insert molding part **56** is fixedly installed at a preset position of the motor coupling unit **20**, the double-layer power terminal part **52** and the double-layer CAN terminal part **55**, which are connected to the first insert molding part **56**, are disposed at intended positions on the first and second motor control circuit units **30** and **40**, as described above.

(73) FIG. **9** is a perspective view schematically illustrating a double-layer sensor terminal part in accordance with an embodiment of the present disclosure.

(74) Referring to FIG. 4, the torque sensor connector **60** in accordance with the embodiment of the present disclosure includes the torque sensor terminal part **61** and a second insert molding part **63**.
(75) The torque sensor terminal part **61** has a structure extended in the x-direction, and is connected to the vehicle-side torque sensor connector **4**, the first motor control circuit unit **30** and the second motor control circuit unit **40**. One side of the torque sensor terminal part **61** in the x-direction is connected to the vehicle-side torque sensor connector **4**, and the other side of the torque sensor terminal part **61** in the x-direction is connected to the first and second motor control circuit units **30** and **40**.

(76) In particular, when the other side of the torque sensor terminal part **61** in the x-direction is extended in the x-direction, the other side of the torque sensor terminal part **61** diverges into a plurality of portions which are connected to the first sensor terminal connection part **34** of the first motor control circuit unit **30** at the first layer and the second sensor terminal connection part **44** of the second motor control circuit unit **40** at the second layer. In the descriptions of the present disclosure, such a part having an end portion diverging into double layers is referred to as a double-layer sensor terminal part **62**.

(77) That is, the torque sensor terminal part **61** in accordance with the present disclosure includes the double-layer sensor terminal part **62** having a bus bar structure that is extended in the x-direction while diverging into a plurality of portions which can be simultaneously connected to the first and second motor control circuit units **30** and **40** disposed in parallel to each other in the x-direction. Furthermore, the torque sensor terminal part **61** has a structure in which a plurality of double-layer sensor terminal parts **62** are aligned in the y-direction and the z-direction. The double-layer sensor terminal part **62** in accordance with the embodiment of the present disclosure includes an external sensor connection part **621**, a first-layer sensor connection part **622**, and a second-layer sensor connection part **626**.

(78) The external sensor connection part **621** is a terminal connected to the vehicle-side torque sensor connector **4**, and the first-layer sensor connection part **622** and the second-layer sensor connection part **626** are terminals connected to the first sensor terminal connection part **34** and the second sensor terminal connection part **44**, respectively. The external sensor connection part **621** is extended in the x-direction, and the first and second-layer sensor connection parts **622** and **626** diverge from the external sensor connection part **621** and extend in the x-direction.

(79) The first-layer sensor connection part **622** diverges so as to protrude in the +x-direction from a +x-direction end portion of the external sensor connection part **621**, and connects to the first sensor terminal connection part **34**. The second-layer sensor connection part **626** diverges so as to protrude in the +z-direction from a +z-direction end of the external sensor connection part **621**, extends in the y-direction, and then extends in the +x-direction.

(80) More specifically, the second-layer sensor connection part **626** diverges so as to protrude in the z-direction and the y-direction from an x-direction middle portion of the external sensor connection part **621**, passes through the terminal through-hole **25** of the motor coupling unit **20** in the +x-direction, and then connects to the second sensor terminal connection part **44**.

(81) Referring to FIG. 9, the second-layer sensor connection part **626** in accordance with the embodiment of the present disclosure includes a sensor diverging part **627**, a sensor middle extension part **628**, and a sensor extension end part **629**.

(82) The sensor diverging part **627** protrudes in the +z-direction from a position corresponding to an x-direction middle portion and the +z-direction end portion of the external sensor connection part **621**. The sensor diverging part **627** protrudes in the z-direction from the other side. The sensor middle extension part **628** is extended in the y-direction from a +z-direction end portion of the sensor diverging part **627**. The sensor extension end part **629** is extended in the +x-direction from a +y-direction end portion of the sensor middle extension part **628**.

(83) The +x-direction end portion of the sensor extension end part **629** is spaced apart in the x-, y- and z-directions from the +x-direction end of the first-layer sensor connection part **622**. That is, the

+x-direction end portions of the first-layer sensor connection part **622** and the second-layer sensor connection part **626** are clearly spaced apart from each other in all directions corresponding to the x-, y- and z-directions.

(84) Therefore, it is possible to lower the possibility that the first-layer sensor connection part **622** and the second-layer sensor connection part **626** will be misassembled or interfere with each other during an assembling or manufacturing process, due to an error or mistake, thereby further improving the productivity. Furthermore, it is possible to minimize the possibility that the damage to one sensor connection part will have an influence on the other sensor connection part, thereby securing structural stability.

(85) The double-layer sensor terminal part **62** having the above-described structure may be easily manufactured as one body through a simple process of cutting a metallic material into a structure including the external sensor connection part **621**, the first-layer sensor connection part **622** and the second-layer sensor connection part **626**, and then bending a portion corresponding to the sensor middle extension part **628** in a multi-stage manner in the +y-direction.

(86) The second insert molding part **63** is made of an insulating material, and the plurality of double-layer sensor terminal parts **62** arranged in the y-direction and the z-direction are connected as one body through insert molding. The plurality of double-layer sensor terminal parts **62** become one part corresponding to the torque sensor connector **60** by the second insert molding part **63**.

(87) The second insert molding part **63** is extended from the external sensor connection part **621** to the second-layer sensor connection part **626**, and fitted and installed into the terminal through-hole **25** of the motor coupling unit **20** in the x-direction. As the second insert molding part **63** is fixedly installed at a preset position of the motor coupling unit **20**, the plurality of double-layer sensor terminal parts **62** connected to the second insert molding part **63** are disposed at intended positions on the first and second motor control circuit units **30** and **40**, as described above.

(88) Like the double-layer power terminal part **52** and the double-layer CAN terminal part **55**, the double-layer sensor terminal part **62** has an x-direction length enough to pass through the first and second circuit boards **31** and **41**, and has a smaller thickness or narrower cross-sectional area than the double-layer power terminal part **52** and the double-layer CAN terminal part **55**. The extension shape and arrangement of the double-layer sensor terminal part **62** may be reliably maintained by the second insert molding part **63**.

(89) Referring to FIG. **2**, the redundancy control unit **70** in accordance the embodiment of the present disclosure includes a circuit detection part **71**, an abnormality determination part **74**, and a driving circuit switching part **75**.

(90) The circuit detection part **71** serves to detect the operation states of the first motor control circuit unit **30**. The circuit detection part **71** in accordance with the embodiment of the present disclosure includes a first circuit detector **72** and a second circuit detector **73**. The first circuit detector **72** serves to detect the operation state of the first motor control circuit unit **30**. The second circuit detector **73** serves to detect the operation state of the second motor control circuit unit **40**.

(91) When an abnormality occurs in the first motor control circuit unit **30** while the first motor control circuit unit **30** is set and operated as the enabled circuit unit, the first circuit detector **72** detects the abnormality. When an abnormality also occurs in the second motor control circuit unit **40** which is not enabled, the second circuit detector **73** detects the abnormality. Such an operation may be performed while power is supplied to both of the first and second motor control circuit units **30** and **40** by the power connector **50** including the double-layer power terminal part **52**.

(92) The abnormality determination part **74** determines whether the first motor control circuit unit **30** is abnormal, according to a signal detected by the circuit detection part **71**. The abnormality determination part **74** may compare the detection signals of the first and second circuit detectors **72** and **73**, and determine that an abnormality occurred in the first or second circuit detector **72** or **73**, when a difference between the signal values is equal to or more than a preset range.

(93) Furthermore, the abnormality determination part **74** may previously secure data on a preset

value for the detection signal of the first circuit detector **72**, for example, a signal value which may occur while the first circuit detector **72** is normally operated, through a plurality of experiments, and compares the data to a preset signal value, in order to determine whether the first circuit detector **72** is abnormal.

(94) When the abnormality determination part **74** determines that an abnormality occurred in the first motor control circuit unit **30** while the motor **5** was controlled by the first motor control circuit unit **30**, the driving circuit switching part **75** stops the operation of the motor **5** by the first motor control circuit unit **30**, and switches the enabled control circuit unit such that the second motor control circuit unit **40** controls the operation of the motor **5**.

(95) That is, when an abnormality occurs in the first motor control circuit unit **30** while the first motor control circuit unit **30** is set and operated as the enabled circuit unit, the driving circuit switching part **75** switches the enabled circuit unit from the first motor control circuit unit **30** to the second motor control circuit unit **40**. Thus, although one of the first and second motor control circuit units **30** and **40** fails, the other one may control the operation of the motor **5**, such that the MDPS for a vehicle can maintain a normal operation state.

(96) FIG. **10** is a cross-sectional view illustrating main parts of an ECU for redundancy of an MDPS for a vehicle in accordance with another embodiment of the present disclosure.

(97) An ECU **1** for redundancy of an MDPS for a vehicle in accordance with the another embodiment of the present disclosure, illustrated in FIG. **10**, is different from the ECU **1** for redundancy of the MDPS for a vehicle in accordance with the embodiment of the present disclosure, illustrated in FIGS. **1** to **9**, in that the first and second motor control circuit units **30** and **40** are not disposed on both sides of the motor coupling unit **20** in the x-direction, but disposed on one side of the motor coupling unit **20** in the x-direction.

(98) Hereafter, while the ECU **1** for redundancy of an MDPS for a vehicle in accordance with the another embodiment of the present disclosure is described, the repeated descriptions of the same, similar or corresponding components as or to those of the ECU **1** for redundancy of an MDPS for a vehicle in accordance with the embodiment of the present disclosure will be omitted, and the following descriptions will be focused on differences therebetween.

(99) The ECU **1** for redundancy of an MDPS for a vehicle in accordance with the another embodiment of the present disclosure has a structure in which the first and second motor control circuit units **30** and **40** are spaced apart from each other with a preset interval provided therebetween in the x-direction, and disposed in parallel to each other only inside the housing **10** or in the space formed between the housing **10** and the motor coupling unit **20**.

(100) Referring to FIG. **10**, the board support part **21** in accordance with the another embodiment of the present disclosure includes a support body part **22**, a contact support part **26**, and a spacing support part **27**.

(101) The support body part **22** is a part forming the basic frame of the board support part **21**, and has a disk shape with widths in the y- and z-directions as a whole. This structure considers that the motor housing **6** has a circular opening, and the shape of the support body part **22** is not specifically limited. The shape of the support body part **22** may be changed to various shapes depending on the shapes and structures of the motor housing **6** and the first and second motor control circuit units **30** and **40**.

(102) The contact support part **26** is formed on one surface of the support body part **22**, facing the housing **10**, i.e. one surface facing the x-direction, such that the second motor control circuit unit **40** abuts on the contact support part **26**. The spacing support part **27** protrudes in the -x-direction from the contact support part **26**, and supports the first motor control circuit unit **30**.

(103) With the first and second motor control circuit units **30** and **40** installed on the contact support part **26** and the spacing support part **27**, respectively, the first and second motor control circuit units **30** and **40** are disposed in parallel to each other in the x-direction with an interval provided therebetween, the interval corresponding to the x-direction height of the spacing support

part 27. The edge of the contact support part 26 is coupled to the housing 10.

(104) The spacing support part 27 is provided as a plurality of spacing support parts which can distributedly support the edge of the first motor control circuit unit 30 from a plurality of points. Furthermore, the spacing support part 27 is formed in the shape of a tube having a -x-direction end to which a pin or bolt member (not illustrated) may be fastened. As the first motor control circuit unit 30 is seated on the spacing support part 27 and fastened through a pin or bolt member, the first motor control circuit unit 30 may be reliably fixed to a preset position inside the housing 10.

(105) As the second motor control circuit unit 40 is disposed so as to abut on the support body part 22, heat generated from the second motor control circuit unit 40 is directly transferred to the support body part 22. Between the respective spacing support parts 27, a path is formed, through which heat discharged from the first motor control circuit unit 30 can smoothly pass. Thus, an excellent heat radiation effect may be stably implemented for both of the first and second motor control circuit units 30 and 40.

(106) In the ECU 1 for redundancy of an MDPS for a vehicle in accordance with the another embodiment of the present disclosure, which has the above-described configuration, when an abnormality occurs in the first motor control circuit unit 30 while the first motor control circuit unit 30 is set and operated as the enabled circuit unit, the redundancy control unit 70 may switch the enabled circuit unit from the first motor control circuit unit 30 to the second motor control circuit unit 40. Thus, although one of the first and second motor control circuit units 30 and 40 fails, the other may control the operation of the motor 5, thereby maintaining the normal operation state of the MDPS for a vehicle.

(107) Furthermore, in accordance with the present disclosure, the redundancy of the MDPS for a vehicle can be implemented by only changing the internal structure of the ECU 1 such that the ECU 1 includes the double-layer power terminal part 52, the double-layer CAN terminal part 55 and the double-layer sensor terminal part 62. Thus, the MDPS can be widely and easily applied to various vehicle specifications, and the numbers of the vehicle-side power connectors 2, the vehicle-side CAN connectors 3 and the vehicle-side torque sensor connectors 4 do not need to be doubled. Therefore, the components may be efficiently disposed in a limited space.

(108) Although exemplary embodiments of the disclosure have been disclosed for illustrative purposes, those skilled in the art will appreciate that various modifications, additions and substitutions are possible, without departing from the scope and spirit of the disclosure as defined in the accompanying claims. Thus, the true technical scope of the disclosure should be defined by the following claims.

Claims

1. An electronic control unit for redundancy of a motor driven power steering for a vehicle, the electronic control unit comprising: a first motor control circuit unit comprising a circuit structure configured to control an operation of a motor; a second motor control circuit unit comprising a circuit structure configured to control the operation of the motor, and disposed in parallel to the first motor control circuit unit along an x-direction; a redundancy control unit configured to control the second motor control circuit unit to perform a motor control, when an abnormality occurs in the first motor control circuit unit; and a double-layer power terminal part comprising a first side connected to a vehicle-side power connector and a second side diverging into double layers, extended in the x-direction, and connected to the first motor control circuit unit and the second motor control circuit unit.

2. The electronic control unit of claim 1, wherein the first motor control circuit unit comprises: a first circuit board configured to independently control the operation of the motor; a first power terminal connection part disposed in a first area of the first circuit board, and connected to a power terminal part connected to a vehicle-side power connector to receive power; a first communication

terminal connection part disposed in a second area of the first circuit board, and connected to a Controller Area Network (CAN) terminal part connected to a vehicle-side CAN connector to receive a signal; and a first sensor terminal connection part disposed in a third area of the first circuit board, and connected to a torque sensor terminal part connected to a vehicle-side torque sensor connector to receive a signal.

3. The electronic control unit of claim 2, wherein the second motor control circuit unit comprises: a second circuit board configured to independently control the operation of the motor; a second power terminal connection part disposed on an x-direction extension line of a first area of the second circuit board, and connected to the power terminal part to receive power; a second communication terminal connection part disposed on an x-direction extension line of a second area of the second circuit board, and connected to the CAN terminal part to receive a signal; and a second sensor terminal connection part disposed on an x-direction extension line of a third area of the second circuit board, and connected to the torque sensor terminal part to receive a signal.

4. The electronic control unit of claim 1, wherein the redundancy control unit comprises: a circuit detection part configured to detect an operation state of the first motor control circuit unit; an abnormality determination part configured to determine whether the first motor control circuit unit is abnormal, according to a signal detected by the circuit detection part; and a driving circuit switching part configured to stop the operation of the motor by the first motor control circuit unit, and perform circuit switching such that the second motor control circuit unit controls the operation of the motor, when the abnormality determination part determines that the first motor control circuit unit is abnormal, while the first motor control circuit unit controls the motor.

5. The electronic control unit of claim 4, wherein the circuit detection part comprises: a first circuit detector configured to detect the operation state of the first motor control circuit unit; and a second circuit detector configured to detect an operation state of the second motor control circuit unit, wherein the abnormality determination part is configured to determine whether the first circuit detector is abnormal by comparing detection signals of the first circuit detector and the second circuit detector or by comparing the detection signal of the first circuit detector to a preset value.

6. The electronic control unit of claim 1, further comprising a motor coupling unit coupled to a motor housing, and disposed between the first motor control circuit unit and the second motor control circuit unit, wherein the motor coupling unit comprises: a board support part fixed to the first motor control circuit unit and the second motor control circuit unit, and configured to receive heat radiated from the first motor control circuit unit and the second motor control circuit unit; and a heat radiation support part connected to the board support part, and configured to radiate heat received from the board support part.

7. The electronic control unit of claim 6, wherein the board support part comprises: a support body part; a first board support part disposed on a first surface of the support body part, such that the first motor control circuit unit abuts the first board support part; a second board support part disposed on a second surface of the support body part, such that the second motor control circuit unit abuts the second board support part; and a terminal through-hole disposed through the support body part, such that a power terminal part connected to a vehicle-side power connector, a Controller Area Network (CAN) terminal part connected to a vehicle-side CAN connector, and a torque sensor terminal part connected to a vehicle-side torque sensor connector are extended through the terminal through-hole in the x-direction.

8. The electronic control unit of claim 6, wherein the board support part comprises: a support body part; a contact support part disposed on a first surface of the support body part, such that one of the first motor control circuit unit and the second motor control circuit unit abuts the contact support part; and a spacing support part protruding from the contact support part in the x-direction, and configured to support the other of the first motor control circuit unit and the second motor control circuit unit.

9. The electronic control unit of claim 1, wherein the double-layer power terminal part comprises:

an external power connection part connected to the vehicle-side power connector; a first-layer power connection part extended from the external power connection part in the x-direction, and connected to one of the first motor control circuit unit and the second motor control circuit unit; and a second-layer power connection part extended in the x-direction from a position spaced apart from the first-layer power connection part, and connected to the other of the first motor control circuit unit and the second motor control circuit unit.

10. The electronic control unit of claim 9, wherein the first-layer power connection part protrudes in the x-direction from one side of the external power connection part, and comprises a first end connected to the first motor control circuit unit, wherein the second-layer power connection part comprises: a power diverging part disposed on the external power connection part to protrude in one of a y-direction and a z-direction; a power middle extension part extended in the other of the y-direction and the z-direction from the power diverging part; and a power extension end part extended in the x-direction from the power middle extension part, and comprising a first end spaced apart from the first end of the first-layer power connection part in the x-direction, the y-direction, and the z-direction.

11. The electronic control unit of claim 9, wherein the double-layer power terminal part comprises: a first bus bar comprising the external power connection part, the first-layer power connection part, and the second-layer power connection part; a second bus bar disposed symmetrically with the first bus bar in a y-direction; and an intersection space defined between the first bus bar and the second bus bar, such that at least a part of a Controller Area Network (CAN) terminal part connected to a vehicle-side CAN connector is disposed in the intersection space.

12. The electronic control unit of claim 1, further comprising a double-layer sensor terminal part comprising a first side connected to a vehicle-side torque sensor connector and a second side diverging into double layers, extended in the x-direction, and connected to the first motor control circuit unit and the second motor control circuit unit, wherein the double-layer sensor terminal part comprises: an external sensor connection part connected to the vehicle-side torque sensor connector; a first-layer sensor connection part diverging in the x-direction from the external sensor connection part, and connected to one of the first motor control circuit unit and second motor control circuit unit; and a second-layer sensor connection part diverging in a y-direction or a z-direction from the external sensor connection part and extended in the x-direction, and connected to the other of the first motor control circuit unit and second motor control circuit unit.

13. The electronic control unit of claim 12, wherein the first-layer sensor connection part is disposed on a first side of the external sensor connection part to extend in the x-direction, and comprises a first end connected to the first motor control circuit unit, wherein the second-layer sensor connection part comprises: a sensor diverging part formed on a second side of the external sensor connection part to protrude in one of the y-direction and the z-direction; a sensor middle extension part extended in the other of the y-direction and the z-direction from the sensor diverging part; and a sensor extension end part extended in the x-direction from the sensor middle extension part, and comprising a first end spaced apart from the first end of the first-layer sensor connection part in the x-direction, the y-direction, and the z-direction.

14. An electronic control unit (ECU) for redundancy of a motor driven power steering for a vehicle for redundancy of a Motor Driven Power Steering (MDPS) for the vehicle, the ECU comprising: a motor coupling unit coupled to a motor; a first motor control circuit unit installed on the motor coupling unit; a second motor control circuit unit installed on the motor coupling unit, and disposed in parallel to the first motor control circuit unit along an x-direction; a power connector comprising a first side connected to a vehicle-side power connector and a second side connected to the first motor control circuit unit and the second motor control circuit unit; and a redundancy control unit configured to control the second motor control circuit unit to perform a motor control, when an abnormality occurs in the first motor control circuit unit, wherein the power connector comprises: a double-layer power terminal part comprising a bus bar structure in which a first side thereof is

connected to the vehicle-side power connector and a second side thereof diverges into double layers, extended in the x-direction, and connected to the first motor control circuit unit and the second motor control circuit unit; a Controller Area Network (CAN) terminal part connected to a vehicle-side CAN connector; and a first insert molding part configured to connect the double-layer power terminal part to the CAN terminal part through insert molding, and installed on the motor coupling part.

15. The electronic control unit of claim 12, wherein the CAN terminal part comprises a double-layer CAN terminal part comprising a bus bar structure in which a first side thereof is connected to the vehicle-side CAN connector and a second side thereof diverges into double layers, extended in the x-direction, and connected to the first motor control circuit unit and the second motor control circuit unit, wherein at least a part of a first area in which the double layer power terminal part is disposed and at least a part of a second area in which the double-layer CAN terminal part is disposed cross or overlap each other.

16. The electronic control unit of claim 14, further comprising a torque sensor connector connected to a vehicle-side torque sensor connector, wherein the torque sensor connector comprises: a double-layer sensor terminal part comprising a bus bar structure in which a first side thereof is connected to the vehicle-side torque sensor connector and a second side thereof diverges into double layers, extended in the x-direction, and connected to the first motor control circuit unit and the second motor control circuit unit; and a second insert molding part formed by insert molding the double-layer sensor terminal part, and fitted and installed into the motor coupling unit in the x-direction.

17. An electronic control unit for redundancy of a motor driven power steering for a vehicle, the electronic control unit comprising: a first motor control circuit unit comprising a circuit structure configured to control an operation of a motor; a second motor control circuit unit comprising a circuit structure configured to control the operation of the motor, and disposed in parallel to the first motor control circuit unit along an x-direction; a redundancy control unit configured to control the second motor control circuit unit to perform a motor control, when an abnormality occurs in the first motor control circuit unit; and a double-layer Controller Area Network (CAN) terminal part comprising a first side connected to a vehicle-side CAN connector and a second side diverging into double layers, extended in the x-direction, and connected to the first motor control circuit unit and the second motor control circuit unit.

18. The electronic control unit of claim 17, wherein the double-layer CAN terminal part comprises: an external CAN connection part connected to the vehicle-side CAN connector; a second-layer CAN connection part extended in the x-direction from the external CAN connection part, and connected to one of the first motor control circuit unit and the second motor control circuit unit; and a first-layer CAN connection part diverging in a y-direction or a z-direction from the external CAN connection part and extended in the x-direction, and connected to the other of the first motor control circuit unit and the second motor control circuit unit.

19. The electronic control unit of claim 18, wherein the second-layer CAN connection part is disposed on a first side of the external CAN connection part to extend in the x-direction, and comprises a first end connected to the second motor control circuit unit, wherein the first-layer CAN connection part comprises: a CAN diverging part formed on a second side of the external CAN connection part to protrude in one of the y-direction and the z-direction; a CAN middle extension part extended in the other of the y-direction and the z-direction from the CAN diverging part; and a CAN extension end part extended in the x-direction from the CAN middle extension part, and comprising a first end spaced apart from the first end of the second-layer CAN connection part in the x-direction, the y-direction, and the z-direction.
